(e) If $u = 6 + \text{ki}$ and $v = 4 + \text{ki}$, find k if $\arg(u.v) = \dfrac{\pi}{4}$.

(e) Describe fully the locus of the points representing z if $\dfrac{z+2\mathrm{i}}{z-2\mathrm{i}}$ is purely imaginary.

(e) Find the values of k for which the equation $6+x-4\sqrt{3x+k}=0$ has no real roots.

(e) Find the equation of the locus described by $|z - 1 + 2i| = |z + 1|$.

(e) The complex number z is given by $z = \dfrac{1+3\mathrm{i}}{p+q\mathrm{i}}$, where p and q are real and $p > q > 0$.

Given that $\mathrm{Arg}(z) = \dfrac{\pi}{4}$, show that $p - 2q = 0$.

(e) α, β, and γ are the three roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$, where a, b, c, and d are real numbers.

(i) Prove that

$$\alpha + \beta + \gamma = \frac{-b}{a}, \quad \alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}, \quad \alpha\beta\gamma = \frac{-d}{a}$$

(ii) Hence prove that $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \frac{bd}{a^2}$

(e) If $x^2 + bx + c$ and $x^2 + dx + e$ have a common factor of $(x - p)$,

prove that $\dfrac{e-c}{b-d} = p$, where b, c, d, e, and p are all real.

(e) Find the Cartesian equation of the locus described by $\arg\left(\frac{z-2}{z+5}\right)=\frac{\pi}{4}$

(e) (i) Find each of the roots of the equation $z^5 - 1 = 0$.

(ii) Let p be the root in part (i) with the smallest positive argument.

Show that the roots in part (i) can be written as $1, p, p^2, p^3, p^4$.

(e) The locus described by $|z-2+3i|=|z-1|$ is a straight line.

Find the gradient of that line.

(e) Find the value of k if the equation $8-x+2\sqrt{2x+k}=0$ has equal roots.

(e) u and v are two complex numbers, such that $|u+v|^2 = |u-v|^2$.

Prove that $u\bar{v}$ is purely imaginary.

(e) Find the Cartesian equation of the locus described by $|z + 2 - 7i| = 2|z - 10 + 2i|$.

Write your answer in the form $(x + A)^2 + (y + B)^2 = K$.

(e) Find all possible values of k that make $u = \dfrac{k+4\mathrm{i}}{1+k\mathrm{i}}$ a purely real number.

(e) z is a complex number such that $z = \dfrac{a + b\,\mathrm{i}}{a - b\,\mathrm{i}}$, where a and b are real numbers.

Prove that $\dfrac{z^2 + 1}{2z} = \dfrac{a^2 - b^2}{a^2 + b^2}$.

(e) If $z = a + b\text{i}$ and $\frac{z}{\overline{z}} = c + d\text{i}$, prove that $c^2 + d^2 = 1$.

(e) The complex number $u = 3 + mi$ is on the locus of points defined by $|z - 8| = |z - 4 + 2i|$.

Find the value of m.

(e) Solve the equation $\dfrac{\sqrt{x+k}+\sqrt{x-k}}{\sqrt{x+k}-\sqrt{x-k}}=4$ for x in terms of k.

(e) Find the values of x and y, given that x and y are real, and $\dfrac{1}{x+\mathrm{i}y}-\dfrac{1}{1+\mathrm{i}}=1-2\mathrm{i}$.

(e) If $|z| = 1$, and $z \neq 1$, prove that $\dfrac{1+z}{1-z}$ is purely imaginary.

(e) Prove that the quadratic equation $x^2 + 3kx + k^2 = 7x + 3k$ will have two distinct real solutions for all real values of k.

(e) Given that $T = \dfrac{a - b\text{i}}{a + b\text{i}}$, where a and b are real constants, prove that $\dfrac{1 + T^2}{2T} = \dfrac{a^2 - b^2}{a^2 + b^2}$.

(e) Find the Cartesian equation of the locus described by $|z + i|^2 + |z - i|^2 = 10$.

Write your solution in the form $x^2 + y^2 = k$.

(e) For complex numbers u and v, prove that if $|u + v| = |u - v|$, then $\frac{u}{v}$ is purely imaginary.

(e) Given that the real part of $\dfrac{z-2\mathrm{i}}{z-4}$ is zero and $z \neq 4$, prove that the locus of points described by z is given by the Cartesian equation $(x-2)^2 + (y-1)^2 = 5$.

(e) If z is a complex number and $|z + 16| = 4|z + 1|$, find the value of $|z|$.

(e) Solve the equation $z^2 = \mathrm{i}(|z|^2 - 4)$.

(e) For complex numbers w and z, prove that:

$$|w+z|^2 - |w-\overline{z}|^2 = 4\,\mathrm{Re}(w)\,\mathrm{Re}(z)$$

where $\mathrm{Re}(w)$ is the real part of w, and $\mathrm{Re}(z)$ is the real part of z.

(e) Find the Cartesian equation of the locus described by $|z+\mathrm{i}|=2|z-5\mathrm{i}|$ in the form $(x-a)^2+(y-b)^2=k^2$.

(e) If $z = a + b\text{i}$ is a non-zero complex number, and $\frac{\text{i}}{z} + \frac{3}{\overline{z}} = 1$, find the values of a and b.

(e) The complex numbers u and v are $u = 3 + \text{i}$ and $v = 1 + 2\text{i}$.

Determine the possible value(s) of the real constant k if $\left|\frac{u}{v} + k\right| = \sqrt{k+2}$.

(e) The straight line with equation $y = mx - 1$, where m is a real constant and $m > 0$, is a tangent to the locus described by $\left|z - 2 + \mathrm{i}\right| = \sqrt{3}$.

Find the Cartesian equation of the locus AND the value of m.

(e) If $z + 2\mathrm{i} = \mathrm{i}z + k$, where k is a real number, and $\dfrac{w}{z} = 2 + 2\mathrm{i}$, where $\mathrm{Im}(w) = 8$, find the value of k.

(e) Consider the complex numbers $u = 2 + 3k\,\mathrm{i}$ and $v = 4 + 5k\,\mathrm{i}$, where k is a real constant.

Show that the complex number $w = \dfrac{u}{v}$ will **not** lie on the line $y = x$ in the Argand diagram, for any value of k.

(e) The locus of a complex number z is described by

$$|z - 1 - 7\mathrm{i}| = 2|z - 4 - 4\mathrm{i}|$$

The complex number $u = 3 + d\mathrm{i}$ lies on this locus.

Find the Cartesian equation of the locus of z, giving your answer in the form $(x - a)^2 + (y - b)^2 = k$ and **also find** the complex number(s) u.

(e) Given that $x+\frac{1}{x}=p$, where p is a real constant, then find the value of $x^3+\frac{1}{x^3}$, giving your answers in terms of p.

(e) The locus of a complex number z is described by $|z - 5i| - |z + 5i| = 4$.

Find the Cartesian equation of the locus of z, giving your answer in the form $ay^2 - bx^2 = k$, where a, b, k are constants.

(e) Given that $\arg\left(\dfrac{d+6\text{i}}{1-d\text{i}}\right)=\dfrac{\pi}{4}$, find the possible value(s) of d, where d is a real constant.

(e) Given that u and v are both complex numbers, solve the following pair of simultaneous equations, giving solutions in the form $a + b\text{i}$.

$$u\text{i} + 2v = 3$$
$$u + (1 - \text{i})v = 4$$

Show your working clearly.